



Q1 String Character Frequency Counter

10 marks

Write a Python program that accepts a string from the user and counts the frequency of each unique character (case-insensitive). Display the results in the format: character → count. Sort the output alphabetically by character.

Constraints:

- Ignore spaces in the count.
- Treat uppercase and lowercase as the same character.
- Sort output alphabetically.

Skills Assessed:

- String iteration, lower(), conditionals
- Dictionary or manual counting
- Sorting using loops or sorted()

Input: Hello World

output:

d → 1

e → 1

h → 1

l → 3

o → 2

r → 1

w → 1

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W → 1

Your Code

```
1 # String Character Frequency Counter
2
3 text = input("Enter a string: ")
4
5 text = text.lower()      # Convert to lowercase
6 freq = {}
7
8 for ch in text:
9     if ch != " ":        # Ignore spaces
10         if ch in freq:
11             freq[ch] += 1
12         else:
13             freq[ch] = 1
14
15 print("Output:")
16 for ch in sorted(freq):
17     print(ch, "->", freq[ch])
```

✓ Submitted

Input

Hello World

Output

```
Enter a string: Output:
d -> 1
e -> 1
h -> 1
l -> 3
o -> 2
```



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Q2 Vowel and Consonant Classifier with Statistics

10 marks

Accept a sentence from the user and display: (a) total vowels and consonants, (b) the most frequent vowel, (c) the least frequent consonant, and (d) the ratio of vowels to consonants as a decimal rounded to 2 places.

Constraints:

- Case-insensitive analysis.
- Ignore spaces and digits.
- Handle sentences with no vowels or no consonants gracefully.

Skills Assessed:

- Character classification with conditionals
- Counting and comparison logic
- Arithmetic and rounding

Sample Input :

Python Programming

Expected Output : Vowels: 5 | Consonants: 12 Most frequent vowel: o (2) Least frequent consonant: y (1) Vowel-to-Consonant Ratio: 0.42



Your Code

```
1 sentence = input("Enter a sentence: ")
2
3 sentence = sentence.lower()
4
5 vowels = "aeiou"
6 vowel_count = 0
7 consonant_count = 0
8
9 vowel_freq = {}
10 consonant_freq = {}
11
12 for ch in sentence:
13     if ch.isalpha():
14         if ch in vowels:
15             vowel_count += 1
16             vowel_freq[ch] = vowel_freq.get(ch, 0) + 1
17         else:
18             consonant_count += 1
19             consonant_freq[ch] = consonant_freq.get(ch, 0) + 1
20
21 most_vowel = max(vowel_freq, key=vowel_freq.get)
22 least_consonant = min(consonant_freq, key=consonant_freq.get)
23
24 ratio = round(vowel_count / consonant_count, 2)
25
26 print("Vowels:", vowel_count)
27 print("Consonants:", consonant_count)
```

Input

Python programming

Output

```
Enter a sentence: Vowels: 4
Consonants: 13
Most frequent vowel: o ( 2 )
Least frequent consonant: y ( 1 )
Vowel-to-Consonant Ratio: 0.31
```




Q3 String Compression and Run-Length Encoding

10 marks

Implement a Run-Length Encoding (RLE) program that: (a) Compresses a string by replacing consecutive repeated characters with character+count. (b) Decompresses the encoded string back to the original. (c) If the compressed string is not shorter, returns the original.

Constraints:

- Single characters without count still output as single char.
- Handle empty strings.
- Decompressor should handle all valid RLE strings.

Skills Assessed:

- String traversal and comparison
- Count accumulation logic
- String building and integer conversion

Sample Input :

aabbbccdddee

Expected Output

Compressed: a2b3c2d4e2

Decompressed: aabbbccdddee

For input 'abc': Compression not beneficial.

Original:



Your Code

```
1 # String Compression and Run-Length Encoding
2
3 text = input("Enter a string: ")
4
5 # Compression
6 compressed = ""
7 count = 1
8
9 for i in range(len(text)):
10     if i < len(text) - 1 and text[i] == text[i + 1]:
11         count += 1
12     else:
13         compressed += text[i]
14         if count > 1:
15             compressed += str(count)
16         count = 1
17
18 # Return original if compressed is not shorter
19 if len(compressed) >= len(text):
20     compressed = text
21
22 print("Compressed:", compressed)
23
24 # Decompression
25 decompressed = ""
26 i = 0
27
```

Input

aabbbbccddddee

Output

```
Enter a string: Compressed:
a2b4c2d4e2
Decompressed: aabbbbccddddee
```



Q4 Advanced Text Statistics Analyser

10 marks

Write a Python program that accepts a paragraph of text and computes: (a) Total words, sentences, and paragraphs. (b) Average word length and average sentence length. (c) The top 5 most frequent words (excluding common stop words: the, is, a, an, in, of, and, to, for). (d) Flesch Reading Ease Score: $206.835 - 1.015 * (\text{words/sentences}) - 84.6 * (\text{syllables/words})$. Approximate syllables as number of vowel groups per word.

Constraints:

- Sentence ends with . ! or ?
- Handle punctuation attached to words.
- Stop words list is hardcoded in the program.

Skills Assessed:

- Complex string parsing and tokenization
- Nested loops for syllable counting
- Floating-point arithmetic and rounding

Sample Input: Python is a powerful programming language

Expected Output :

Words: 45 | Sentences: 4 | Paragraphs: 1

Avg Word Length: 5.2 | Avg Sentence Length: 11.25

Top 5 Words: python(3), programming(2), language(2), data(2), code(1)

Flesch Reading Ease: 58.3 (Standard)

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Your Code

```
1 import re
2
3 text = input("Enter a paragraph: ")
4
5 # Total words
6 words = re.findall(r'\b\w+\b', text.lower())
7 total_words = len(words)
8
9 # Total sentences
10 sentences = re.split(r'[.!?]+', text)
11 sentences = [s for s in sentences if s.strip()]
12 total_sentences = len(sentences)
13
14 # Total paragraphs
15 paragraphs = [p for p in text.split("\n") if p.strip()]
16 total_paragraphs = len(paragraphs)
17
18 # Average word length
19 avg_word_len = round(sum(len(w) for w in words) / total_words, 2)
20
21 # Average sentence length
22 avg_sent_len = round(total_words / total_sentences, 2)
23
24 # Stop words
25 stop = ["the", "is", "a", "an", "in", "of", "and", "to", "for"]
26
27 freq = {}
```

Input

Python is powerful programming
languages

Output

```
Enter a paragraph: Total Words: 5
Total Sentences: 1
Total Paragraphs: 1
Average Word Length: 7.2
Average Sentence Length: 5.0
Top 5 Frequent Words:
```